# HAONAN CHEN

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## EXPERIENCES

**Postdoctoral Research Fellow**
Harvard University 09/2025 – Present
Advisor: Yilun Du

## EDUCATION

**Ph.D. in Electrical and Computer Engineering**
University of Illinois Urbana-Champaign 08/2020 - 08/2025

**B.Sc. in Electrical Engineering with Highest Honors**
University of Illinois Urbana-Champaign 09/2016 - 05/2020
GPA: 3.98/4.0

**B.Eng. in Electrical Engineering and Automation**
Zhejiang University 09/2016 - 06/2020
GPA: 3.99/4.0 (Rank: 1/30)

## JOURNAL PUBLICATIONS

**[RA-L'26]** C. Liu, H. Chen, S. H. Høeg*, S. Yao*, Y. Li, K. Hauser, Y. Du. "Flexible Multitask Learning with Factorized Diffusion Policy." *IEEE Robotics and Automation Letters,* 2026. [pdf] [Website]

**[ITSM'24]** A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Lessons in Cooperation: A Qualitative Analysis of Driver Sentiments towards real-time Advisory Systems through a Focus Group User Study." *IEEE Intelligent Transportation Systems Magazine,* 2024. [pdf]

**[JATS'24]** A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Cooperative Advisory Residual Policies for Congestion Mitigation." *ACM Journal on Autonomous Transportation Systems*, 2024. [pdf]

## PREPRINTS

Y. Ma*, M. Tippur*, P. Ye, S. Q. Liu, H. Chen, F. R. Cottrell, E. Adelson. "GlowTact: Simple and Compact Vision-Based Tactile Sensing with High Sensitivity and Spatial Resolution." *arXiv preprint,* 2026. [pdf] [Website]

R. Sathua, H. Chen, K. Driggs-Campbell. "ARGUS: Aligning Robot Scene Geometry Under Shifting Views with Large 3D Vision Models." *arXiv preprint,* 2026. [pdf] [Website]

S. Lu, H. Chen, W. Ye, Y. Du. "DriftWorld: Fast World Modeling through Drifting." *arXiv preprint,* 2026. [pdf] [Website] [Code]

X. Han*, H. Xiong*, H. Chen, C. Liu, A. Torralba, Y. Zhu, Y. Du. "B-spline Policy: Accelerating Manipulation Policies via B-spline Action Representations." *arXiv preprint,* 2026. [pdf] [Website] [Code]

P. Ye, Y. Ma, H. Chen, G. Wang, Y. Jing, B. Sheil, Y. Du, E. Adelson. "You Only Touch Once: 6-DoF Object Pose Estimation from Single Tactile Contact." *arXiv preprint,* 2026. [pdf]

J. Gao, C. Liu, P. Wu, H. Chen, Y. Du. "IMPACT: Learning Internal-Model Predictive Control for Forceful Robotic Manipulation." *arXiv preprint,* 2026. [pdf] [Website] [Code]

## PEER REVIEWED CONFERENCE PUBLICATIONS

**[NeurIPS'26]** H. Alzayer, W. Huang, H. Chen, C. Luey, L. Zhang, M. Agrawala, G. Wetzstein, L. Fei-Fei, Y. Du, J. Wu, J.-B. Huang. "Masked Visual Actions for Unified World Modeling." *Advances in Neural Information Processing Systems,* 2026. [pdf] [Website] [Code]

**[CoRL'26]** H. Chen*, Y. Ma*, S. Tian, X. Han, W. Huang, F. Wu, Y. Li, J. Wu, E. H. Adelson, Y. Du. "CoStream: Composing Simple Behaviors for Generalizable Complex Manipulation." *Conference on Robot Learning,* 2026. [pdf] [Website]

**[ECCV'26]** H. Chen, K. Zhou, H. Hua, K. Zhang, J. Qian, W. Ma, H. Chen, C. Liu, Y. Zhao, X. Wang, W. Li, A. Yuille, P. P. Liang, Y. Du. "MemoBench: Benchmarking World Modeling in Dynamically Changing Environments." *European Conference on Computer Vision,* 2026. [pdf] [Website] [Code]

**[RSS'26]** C. Liu, X. Han, J. Gao, Y. Zhao, H. Chen, Y. Du. "OAT: Ordered Action Tokenization." *Robotics: Science and Systems,* 2026. [pdf] [Website] [Code]
Finalist - Best/Outstanding Paper Awards

**[CVPR'26]** H. Liu*, S. Yao*, H. Chen, J. Gao, J. Mao, J.-B. Huang, Y. Du. "SIMPACT: Simulation-Enabled Action Planning using Vision-Language Models." *IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 2026. [pdf] [Website]

**[ICRA'26]** H. Chen, J. Xu*, H. Chen*, K. Hong, B. Huang, C. Liu, J. Mao, Y. Li, Y. Du+, K. Driggs-Campbell+. "Multi-Modal Manipulation via Multi-Modal Policy Consensus." *IEEE International Conference on Robotics and Automation,* 2026. [pdf] [Website] [Code]
Best Paper Award at [CVPR 2026 Workshop on Multi-Sensory Modeling for Embodied Intelligence]
Featured in Video Friday on [IEEE Spectrum]

**[CoRL'25]** H. Chen, C. Zhu, S. Liu, Y. Li, K. Driggs-Campbell. "Tool-as-Interface: Learning Robot Policies from Observing Human Tool Use." *Conference on Robot Learning,* 2025. [pdf] [Website] [Code]
Best Paper Award at [ICRA 2025 workshop on Foundation Models and Neural-Symbolic (NeSy) AI for Robotics]
Best Presentation Award at [CSL Student Conference 2025]

**[RSS'25]** S. Peng, H. Chen, K. Driggs-Campbell. "Towards Uncertainty Unification: A Case Study for Preference Learning." *Robotics: Science and Systems,* 2025. [pdf] [Website] [Code]
Best Paper Award at [ICRA 2026 Workshop on Uncertainty in Open-World Robotics]

**[ICRA'25]** H. Chen, J. Xu*, L. Sheng*, T. Ji, S. Liu, Y. Li, K. Driggs-Campbell. "Learning Coordinated Bimanual Manipulation Policies using State Diffusion and Inverse Dynamics Models." *IEEE International Conference on Robotics and Automation,* 2025. [pdf] [Website]

**[CoRL'23]** H. Chen, Y. Niu*, K. Hong*, S. Liu, Y. Wang, Y. Li, K. Driggs-Campbell. "Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks." *Conference on Robot Learning,* 2023. [pdf] [Website]
Finalist - Best Paper/Best Student Paper Awards

**[CASE'23]** Z. Huang, Y. Mun, H. Chen, Y. Xie, Y. Niu, X. Li, N. Zhong, H. You, D.L. McPherson, K. Driggs-Campbell. "Towards Safety of Multi-Level Human-Robot Interaction in Industrial Tasks." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023. [pdf]

**[CASE'23]** H. Chen*, Y. Mun*, Z. Huang, Y. Niu, Y. Xie, D.L. McPherson, K. Driggs-Campbell. "Learning Task Skills and Goals Simultaneously from Physical Interaction." *IEEE Conference on Automation*

*Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023. [pdf]

**[ITSC'23]** A. Hasan, N. Chakraborty*, H. Chen*, J-H. Cho, C. Wu, K. Driggs-Campbell. "PeRP: Personalized Residual Policies For Congestion Mitigation Through Co-operative Advisory Systems." *IEEE International Conference on Intelligent Transportation Systems,* 2023. [pdf] [Website]
Best Presentation Award at [CSL Student Conference 2024]

**[ITSC'22]** H. Chen, T. Ji, S. Liu, K. Driggs-Campbell. "Combining Model-Based Controllers and Generative Adversarial Imitation Learning for Traffic Simulation." *IEEE International Conference on Intelligent Transportation Systems (ITSC 2022).* [pdf]

**[ICRA'22]** S. Liu, P. Chang, H. Chen, N. Chakraborty, K. Driggs-Campbell. "Learning to Navigate Intersections with Unsupervised Driver Trait Inference." *IEEE International Conference on Robotics and Automation,* 2022. [pdf] [Website]

**[IROS'20]** P. Chang, S. Liu, H. Chen, K. Driggs-Campbell. "Robot Sound Interpretation: Combining Sight and Sound in Learning-Based Control." *IEEE/RSJ International Conference on Intelligent Robots and Systems,* 2020. [pdf] [Video]

## INVITED TALKS & PANELS

**July 2026** Professor Zackory Erickson's Group, Carnegie Mellon University

**April 2026** Professor Paul Liang's Group, MIT
Title: *Multi-Modal Action Composition for Generalizable Robot Manipulation*

**October 2025** Computational Robotics Lab, Harvard University
Title: *Multi-Modal Action Composition for Generalizable Robot Manipulation*

**June 2025** Tacta Systems
Title: *Learning What, How, and Where from Humans: Structured Inductive Biases for Robot Learning*

**June 2025** Professor Jiajun Wu's Lab, Stanford University
Title: *Learning What, How, and Where from Humans: Structured Inductive Biases for Robot Learning*

**May 2025** Professor Roberto Martín-Martín's Lab, University of Texas at Austin
Title: *Toward Human-Robot Alignment through Human Behavior Priors*

**May 2025** St. Matthew Catholic School (Outreach Talk to Grade 5 Students)
Title: *Toward Human-Robot Alignment through Human Behavior Priors*

**April 2025** Professor Anima Anandkumar 's Lab, Caltech
Title: *Deployable Robot Learning from Human Priors*

**April 2025** Professor Katerina Fragkiadaki's Lab, Carnegie Mellon University
Title: *Deployable Robot Learning from Human Priors*

**March 2025** Safe AI Lab, Carnegie Mellon University
Title: *Building Deployable and Pragmatic Robots from Human Priors*

**March 2025** Parasol Lab, University of Illinois Urbana-Champaign
Title: *Tool-as-Interface: Learning Robot Tool Use from Human Play through Imitation Learning*

## MEDIA COVERAGE

- **Multi-Modal Manipulation via Multi-Modal Policy Consensus** — Featured in Video Friday, IEEE Spectrum
- **OAT: Ordered Action Tokenization** — MarkTechPost

- **Tool-as-Interface** — Featured in Video Friday, IEEE Spectrum; The Grainger College of Engineering; TechXplore; Hackster.io

## TEACHING EXPERIENCE

### Graduate Teaching Assistant

- **UIUC ECE 484:** Safe Autonomy — **Spring 2025**
  Instructor: Prof. Sayan Mitra

### Undergraduate Teaching Assistant

- **ZJU-UIUC Math 241:** Calculus III — **Fall 2018**
  Instructor: Prof. Thomas Honold
- **ZJU-UIUC ECE 310:** Digital Signal Processing
  **ZJU-UIUC ECE 311:** Digital Signal Processing Lab — **Fall 2019**
  Instructor: Prof. Mark D. Butala

## MENTORED STUDENTS

### Undergraduate Students

- Jiaming Xu — Placement: University of Illinois Urbana-Champaign
- Lily Sheng — Placement: Tsinghua University
- Yilong Niu — Placement: University of Illinois Urbana-Champaign
- Cheng Zhu — Placement: University of Pennsylvania

## AWARDS AND HONORS

**Finalist - Best/Outstanding Paper Awards**, Robotics: Science and Systems — 2026
**Conference Travel Award**, Graduate College, University of Illinois Urbana-Champaign — 2023
**Finalist - Best Paper/Best Student Paper Awards**, Conference on Robot Learning — 2023
**Bronze Tablet Award**, University of Illinois Urbana-Champaign — 2020
**Excellent Graduate**, Zhejiang University — 2020
**Student Speaker at Graduation**, ZJU-UIUC Institute — 2020
**Provincial Government Scholarship**, Zhejiang Province, China (Top 3%) — *2019*
**Dean's List**, University of Illinois Urbana-Champaign — 2019
**National Scholarship**, China (Top 1.8%) — 2018
**Excellent Undergraduate Scholarship, Zhejiang University** (Top 3%) — 2018-2019
**Academic Excellence Scholarship Award**, ZJU-UIUC Institute — 2017-2019
**Top Ten Social Practice**, Zhejiang University (Top 1.5%) — 2017

## SERVICE AND LEADERSHIP

- **IROS 2026 Workshop on Compositional and Modular Learning in the Era of Scaling in Robotics**
  Co-organizer · 2026
  Workshop Website
- **CoRL 2026 Workshop on Bimanual Dexterous Tool Use: From Contact-Rich Execution to Generalist Robot Intelligence**
  Co-organizer · 2026
  Workshop Website

- **CoRL 2026 Workshop on Compositional Generalization for Real-World Robot Learning**
  Co-organizer · 2026
  Workshop Website

- **ICRA 2025 Workshop on Learning Meets Model-Based Methods for Contact-Rich Manipulation** Lead Organizer · May 2025
  Workshop Website
  Successfully Raised $1,400 in sponsorship from three companies and three IEEE RAS Technical Committees. The workshop attracted over 300 attendees.

- **Robotics Seminar @ Illinois** Student Co-Organizer · August 2023–May 2025
  Seminar Series

- **Coordinated Science Laboratory Student Conference** Reception Chair · 2024
  Conference Website

## PROFESSIONAL SERVICE

- **Session Chair:**
  - IEEE International Conference on Robotics and Automation (ICRA), 2025 – Language-Guided Manipulation
  - CASE - IEEE Conference on Automation Science and Engineering, 2023
- **Reviewer:**
  - IEEE RAL - IEEE Robotics and Automation Letters
  - CoRL - Conference on Robot Learning
  - ICLR - International Conference on Learning Representations
  - ICRA - IEEE International Conference on Robotics and Automation
  - ITSC - IEEE International Conference on Intelligent Transportation Systems
  - CASE - IEEE Conference on Automation Science and Engineering

## OUTREACH AND EXHIBITIONS

**Sweet Gestures: Interactive Cupcake Sprinkling Robot** 2024

- **Engineering Open House 2024** (Urbana-Champaign, IL)
  Designed and showcased robotic exhibit for 20,000+ attendees, engaging K-12 students from across Illinois
- **Society of Women Engineers WE24 Conference** (Chicago, IL)
  Demonstrated interactive robotics platform to 11,000+ conference participants

**STEM Talk: Toward Human-Robot Alignment through Human Behavior Priors** May 2025
Presented to Grade 5 students at St. Matthew Catholic School, fostering early interest in robotics and AI through accessible storytelling.

## EXTRA-CURRICULAR SERVICE

Founding Team Leader, Robotic Team for RoboMaster Challenge, ZJU-UIUC Institute 10/2017-06/2018